MSK Anatomy (Upper & Lower Limb) — NBME Study Guide

Back & shoulder · Brachial plexus · Arm/forearm/hand · Nerve injuries · Vessels · Gluteal region & hip · Thigh · Knee · Leg · Ankle & foot · Lower limb nerves

Red = extra high yield (classic NBME) Black = core content *Italic blue = mnemonic*

1. Back, shoulder & scapular region

Muscle	Nerve	Action / lesion
Trapezius	**Spinal accessory (CN XI)**	**Elevates/retracts/rotates scapula up; CN XI injury (posterior triangle surgery/node biopsy) → shoulder droop, can't shrug or abduct above 90°**
Latissimus dorsi	Thoracodorsal (C6–8)	Extends, adducts, medially rotates arm
Levator scapulae, rhomboids	Dorsal scapular (C5)	Elevate/retract scapula; lesion → mild lateral scapular winging
Serratus anterior	**Long thoracic (C5–7)**	**Protracts, upward rotates scapula (abduction >90°); lesion (mastectomy/axillary dissection, stab) → medial winged scapula pushing against wall**
Deltoid	**Axillary (C5–6)**	**Abducts 15–90°; lesion → flat shoulder, ↓ abduction, numb "regimental badge" area**
Supraspinatus	**Suprascapular**	**Initiates abduction (0–15°); most commonly torn rotator cuff; impingement under acromion; empty-can test, painful arc**
Infraspinatus	Suprascapular	Lateral rotation (pitching injury)
Teres minor	Axillary	Lateral rotation, adduction
Subscapularis	Upper & lower subscapular	Medial rotation, adduction (lift-off test)
Teres major	Lower subscapular	Medial rotation, adduction, extension

- **Rotator cuff = SITS** (Supraspinatus, Infraspinatus, Teres minor, Subscapularis); all insert on greater tubercle except subscapularis (lesser).
- **Quadrangular space** (teres minor, teres major, long head triceps, humerus): **axillary nerve + posterior circumflex humeral artery**. Triangular space: circumflex scapular artery. **Triangular interval**: radial nerve + profunda brachii.
- **Shoulder dislocation is usually anterior (inferior)** → axillary nerve at risk. Posterior dislocation: seizures/electrocution.
- Clavicle fracture (middle third): medial fragment up (SCM), lateral fragment down (arm weight) and in (pec major). Triangle of auscultation: trapezius, latissimus, scapula.
- Scapular anastomosis (subclavian ↔ axillary) bypasses axillary/subclavian obstruction.

2. Brachial plexus & spinal levels

- **Roots C5–T1 → Trunks (upper C5–6, middle C7, lower C8–T1) → Divisions (anterior/posterior) → Cords (lateral, posterior, medial, named relative to axillary artery) → Branches** *("Randy Travis Drinks Cold Beer")*.
- Roots give **dorsal scapular** (C5) and **long thoracic** (C5–7); upper trunk gives **suprascapular** and nerve to subclavius.
- Lateral cord → musculocutaneous, lateral pectoral, lateral root of median. Medial cord → ulnar, medial pectoral, medial cutaneous nerves, medial root of median. **Posterior cord → axillary, radial, upper/lower subscapular, thoracodorsal** *(STAR: Subscapulars, Thoracodorsal, Axillary, Radial)*.

Injury	Cause	Roots	Picture

Erb palsy (upper trunk)	**Infant lateral neck traction in delivery; adult fall on shoulder**	**C5–C6**	**“Waiter’s tip”: arm adducted (deltoid, supraspinatus), medially rotated (infraspinatus), elbow extended (biceps/brachialis), forearm pronated**
Klumpke palsy (lower trunk)	**Upward arm traction (grabbing branch, delivery with arm up); Pancoast tumor, cervical rib**	**C8–T1**	**Total claw hand (intrinsic loss), ± Horner syndrome (T1 sympathetics)**
Thoracic outlet syndrome	Cervical rib, scalene compression	Lower trunk + subclavian vessels	Hand intrinsic atrophy, ischemia, pain

Reflex	Root
Biceps	**C5 (C5–6)**
Brachioradialis	**C6**
Triceps	**C7**
Patellar	**L4 (L3–4)**
Achilles	**S1**

Reflexes count up: S1–2 ankle, L3–4 knee, C5–6 biceps, C7–8 triceps. Dermatomes: C6 thumb, C7 middle finger, C8 little finger, T4 nipple, T10 umbilicus, L4 knee/medial leg, L5 dorsum of foot, S1 lateral foot/sole.

3. Arm, forearm & elbow

- **Anterior arm** (musculocutaneous): biceps (supination > flexion; radial tuberosity), brachialis (main elbow flexor), coracobrachialis. **Posterior arm** (radial): triceps, anconeus.
- **Anterior forearm** (median, except FCU and medial half FDP = ulnar): pronators, wrist & finger flexors from **medial epicondyle** (golfer’s elbow). **Posterior forearm** (radial/PIN): extensors from **lateral epicondyle** (**tennis elbow = ECRB**). Brachioradialis = radial nerve but flexes elbow (midprone).
- **Supination: biceps (most powerful) + supinator. Pronation: pronator teres + pronator quadratus.**
- **Cubital fossa (lateral → medial): biceps tendon, brachial artery, median nerve** *(“TAN”: Tendon, Artery, Nerve; radial nerve lateral)*. Median cubital vein superficial (venipuncture). Boundaries: brachioradialis (lateral), pronator teres (medial).
- **Supracondylar humerus fracture** (children, FOOSH): brachial artery + median nerve injury → Volkmann ischemic contracture (compartment syndrome).
- **Nursemaid’s elbow**: radial head subluxation out of the annular ligament (pulling a child’s arm).
- Elbow joint: humeroulnar (hinge), humeroradial; proximal radioulnar (pivot). UCL (Tommy John) resists valgus.

4. Hand & wrist

- Carpal bones: *“So Long To Pinky, Here Comes The Thumb”* — proximal row lateral → medial: Scaphoid, Lunate, Triquetrum, Pisiform; distal row medial → lateral: Hamate, Capitate, Trapezoid, Trapezium.
- **Scaphoid = most commonly fractured carpal** (FOOSH; tender in anatomical snuffbox; may be X-ray occult) → **avascular necrosis** of proximal pole (retrograde blood supply) and nonunion. **Lunate = most commonly dislocated** (anteriorly) → acute carpal tunnel syndrome. Hook of hamate fracture → ulnar nerve (Guyon canal).
- **Carpal tunnel** (flexor retinaculum roof): median nerve + 9 tendons (4 FDS, 4 FDP, FPL). FCR not inside (separate). Compression (pregnancy, RA, hypothyroid, diabetes, amyloid, repetitive use) → numb lateral 3½ digits, **thenar atrophy**, Tinel/Phalen positive; **palm sensation spared** (palmar cutaneous branch passes superficial to retinaculum).
- **Guyon canal** (pisiform & hook of hamate): ulnar nerve + artery (cyclists).
- **Anatomical snuffbox**: lateral = APL & EPB; medial = EPL; floor = scaphoid & trapezium; contains **radial artery** (and cephalic vein origin, superficial radial nerve over roof).
- **Thenar** (median recurrent branch): APB, FPB (superficial head), opponens pollicis. **Hypothenar** (ulnar): ADM, FDMB, ODM. **Adductor pollicis**, all interossei, lumbricals 3–4 = **ulnar**; lumbricals 1–2 = median.
- **Interossei: PAD (palmar adduct), DAB (dorsal abduct)**; lumbricals flex MCP + extend IP. Card test (palmar interossei, ulnar).

- Extensor compartments (6): APL/EPB (de Quervain tenosynovitis, Finkelstein test) | ECRL/ECRB | EPL (around Lister tubercle) | ED/EI | EDM | ECU.
- Deformities: mallet finger (terminal extensor), boutonnière (central slip), swan-neck (RA), Dupuytren (palmar fascia), jersey finger (FDP avulsion, ring finger), gamekeeper/skier thumb (UCL of thumb MCP).

5. Upper limb nerve injuries — MUST KNOW

Nerve	Injured by	Motor loss / sign	Sensory loss
Axillary (C5–6)	**Surgical neck fracture, anterior shoulder dislocation**	**Deltoid (abduction 15–90°), teres minor; flat shoulder**	**Lateral shoulder ("regimental badge")**
Musculocutaneous (C5–7)	**Upper trunk compression**	**Biceps, brachialis, coracobrachialis → weak elbow flexion/supination**	**Lateral forearm**
Radial (C5–T1)	**Midshaft humerus fracture (radial groove), "Saturday night palsy", crutches (axilla), radial head fracture (PIN)**	**Wrist drop, ↓ finger/thumb extension; axillary lesion also triceps loss; midshaft spares triceps. PIN: finger drop, wrist extends with radial deviation (ECRL spared), no sensory loss**	**Posterior arm/forearm, dorsal lateral hand & first dorsal web space**
Median (C5–T1)	**Supracondylar fracture (proximal), carpal tunnel/wrist laceration (distal)**	**Proximal: "hand of benediction" / OK sign when making a fist (index & middle can't flex). Distal: ape hand, thenar atrophy, ↓ opposition. AIN: can't make OK sign, no sensory loss**	**Palmar lateral 3½ digits (+ thenar palm if proximal)**
Ulnar (C8–T1)	**Medial epicondyle fracture ("funny bone"), Guyon canal (hook of hamate, cyclists)**	**Ulnar claw (4th, 5th: MCP hyperextension, IP flexion) — worse with distal lesion (ulnar paradox: proximal lesion also paralyzes FDP to 4–5, less clawing); ↓ abduction/adduction of fingers; Froment sign (adductor pollicis); hypothenar & 1st dorsal interosseous wasting**	**Medial 1½ digits, hypothenar**
Long thoracic (C5–7)	**Mastectomy, axillary node dissection, stab**	**Winged scapula (medial), can't abduct >90°**	**None**
Recurrent branch of median	Superficial laceration of thenar eminence	Thenar paralysis, ↓ opposition	None

Claw hand: lumbricals lost → MCP extended, IP flexed. "Hand of benediction" appears when trying to make a fist (proximal median); ulnar claw appears at rest (distal ulnar).

6. Breast & axilla

- **~75% of breast lymph → axillary nodes** (pectoral/anterior first; sentinel node); medial breast → parasternal (internal thoracic) nodes. Suspensory (Cooper) ligaments → skin dimpling; lymphatic blockage → peau d'orange.
- Axilla walls: anterior (pec major/minor), posterior (subscapularis, teres major, latissimus), medial (serratus anterior, ribs), lateral (intertubercular groove). Contents: axillary artery/vein, brachial plexus cords, nodes.
- **Axillary artery divided by pectoralis minor into 3 parts**: 1st part: superior thoracic (1 branch); 2nd: thoracoacromial, lateral thoracic (2); 3rd: subscapular, anterior & posterior circumflex humeral (3) *("Screw The Lawyer Save A Patient")*.
- Axillary dissection risks: **long thoracic** (winging), **thoracodorsal** (weak latissimus), intercostobrachial (numb medial arm), lymphedema.

7. Upper limb blood supply

- Subclavian → (1st rib lateral border) axillary → (teres major inferior border) **brachial** → (cubital fossa) radial + ulnar.
- **Profunda brachii runs with the radial nerve in the radial groove** (midshaft fractures). Posterior circumflex humeral with axillary nerve (surgical neck).

- **Superficial palmar arch = mainly ulnar artery; deep palmar arch = mainly radial artery.** **Allen test** checks collateral flow before radial artery puncture/cannulation.
- Veins: cephalic (lateral, deltopectoral groove → axillary), basilic (medial → brachial/axillary), median cubital (connects; venipuncture).
- BP cuff over brachial artery; pulses: radial at wrist lateral to FCR tendon.

8. Gluteal region & hip joint

Nerve	Muscles	Lesion
Superior gluteal (L4–S1)	**Gluteus medius, minimus, TFL (abduct, medially rotate thigh; keep pelvis level)**	**Trendelenburg sign: pelvis drops on the opposite (unsupported) side when standing on the affected limb; waddling gait. Cause: IM injection, hip surgery**
Inferior gluteal (L5–S2)	**Gluteus maximus (extension, lateral rotation; climbing stairs, rising from a chair)**	**Difficulty climbing stairs, rising from seated; posterior lurch**

- **Piriformis = key landmark**: superior gluteal nerve/vessels exit **above** it; sciatic, inferior gluteal, pudendal nerves, posterior femoral cutaneous exit **below** it (greater sciatic foramen). Pudendal re-enters via lesser sciatic foramen (block near ischial spine).
- Lateral rotators: piriformis, superior/inferior gemelli, obturator internus (nerve to obturator internus), quadratus femoris, obturator externus (obturator n.).
- **IM injection: superolateral quadrant of buttock (or ventrogluteal) to avoid sciatic nerve.**
- Hip joint: ball-and-socket; **iliofemoral ligament (Y of Bigelow) strongest, prevents hyperextension**; pubofemoral, ischiofemoral; ligament of head (artery of obturator origin, important in children).
- **Femoral head blood supply mainly medial circumflex femoral artery** (retinacular branches) → femoral neck fracture → **avascular necrosis**. Displaced fracture: limb shortened, laterally rotated.
- **Posterior hip dislocation** (dashboard injury, most common): limb shortened, **flexed, adducted, medially rotated**; sciatic nerve risk. Anterior: abducted, laterally rotated.
- Pediatric hip: DDH (Ortolani/Barlow), Legg–Calvé–Perthes (AVN, 4–10 y), SCFE (obese adolescent, knee pain, limp).

9. Thigh, femoral triangle & adductor canal

- Anterior compartment (femoral n.): quadriceps (knee extension; rectus femoris also flexes hip), sartorius, iliopsoas (strongest hip flexor). Medial (obturator n.): adductors, gracilis, (pectineus femoral). Posterior (tibial division of sciatic): hamstrings — semitendinosus, semimembranosus, biceps femoris long head (short head = common fibular).
- **Femoral triangle: inguinal ligament (superior), sartorius (lateral), adductor longus (medial). Contents lateral → medial: Nerve, Artery, Vein, (Empty space), Lymphatics** *(NAVEL)*. **Femoral sheath** contains artery, vein and femoral canal (lymphatics) — NOT the nerve. **Femoral hernia** through femoral canal, below and lateral to pubic tubercle, more common in women, high strangulation risk.
- Femoral artery pulse at mid-inguinal point; access for catheterization.
- **Adductor (Hunter) canal**: femoral artery, femoral vein, **saphenous nerve**, nerve to vastus medialis; roof = sartorius. Femoral vessels pass through the **adductor hiatus** → popliteal.
- Pes anserinus (medial tibia): Sartorius, Gracilis, semiTendinosus *("Say Grace before Tea")*.

10. Knee joint & popliteal fossa

Structure	Function / injury	Test
ACL (lateral femoral condyle → anterior tibia)	**Prevents anterior tibial translation; non-contact pivot/twist**	**Anterior drawer, Lachman**
PCL (medial condyle → posterior tibia)	**Prevents posterior translation; dashboard injury**	**Posterior drawer, sag sign**
MCL	**Resists valgus (blow to lateral knee); attached to medial meniscus**	**Valgus stress**
LCL	Resists varus; not attached to meniscus	Varus stress

Medial meniscus	**C-shaped, less mobile → injured more**	**McMurray; locking, joint-line pain**

- **"Unhappy triad": ACL + MCL + meniscus** (lateral blow to planted knee; classically medial meniscus, though lateral is more common acutely).
- Screw-home/locking; **popliteus unlocks** the knee (laterally rotates femur). Prepatellar bursitis ("housemaid's knee"); Baker cyst (semimembranosus–gastrocnemius bursa).
- **Popliteal fossa**: borders biceps femoris (superolateral), semimembranosus/semitendinosus (superomedial), gastrocnemius heads (inferior). Contents superficial → deep: **tibial nerve, popliteal vein, popliteal artery** (deepest; aneurysm, injury in knee dislocation). Common fibular nerve along biceps tendon.
- Patellar dislocation usually lateral (vastus medialis oblique resists). Osgood–Schlatter = tibial tuberosity apophysitis.

11. Leg compartments

Compartment	Muscles	Nerve	Artery	Action
Anterior	**Tibialis anterior, EHL, EDL, fibularis tertius**	**Deep fibular**	**Anterior tibial (→ dorsalis pedis)**	**Dorsiflexion, toe extension, inversion (TA)**
Lateral	**Fibularis longus, brevis**	**Superficial fibular**	**(Fibular artery branches)**	**Eversion**
Posterior superficial	**Gastrocnemius, soleus, plantaris**	**Tibial**	**Posterior tibial**	**Plantarflexion**
Posterior deep	**Tibialis posterior, FDL, FHL, popliteus**	**Tibial**	**Posterior tibial, fibular**	**Plantarflexion, inversion, toe flexion**

- **Anterior compartment syndrome** (tibial fracture, crush, reperfusion): pain out of proportion, pain on passive stretch, tense compartment; late pulselessness → fasciotomy. Shin splints = medial tibial stress.
- **Behind medial malleolus (tarsal tunnel): "Tom, Dick And Very Nervous Harry"** — Tibialis posterior, flexor Digitorum longus, posterior tibial Artery, Vein, tibial Nerve, flexor Hallucis longus. Pulse: posterior tibial artery here; dorsalis pedis lateral to EHL tendon on dorsum.
- Achilles (calcaneal) tendon rupture: pop, can't plantarflex strongly; **Thompson test** positive.

12. Ankle & foot

- Ankle (talocrural) = hinge: dorsiflexion/plantarflexion; **most stable in dorsiflexion** (wider anterior talus). Inversion/eversion at **subtalar** and transverse tarsal joints.
- **Inversion sprain (most common): anterior talofibular ligament (ATFL) tears first**, then calcaneofibular. Deltoid ligament (medial) resists eversion — so strong that eversion injuries avulse the medial malleolus.
- Arches: medial longitudinal (keystone talus head/navicular; spring ligament, tibialis posterior, plantar aponeurosis), lateral longitudinal, transverse (fibularis longus).
- Plantar fasciitis (heel pain worst first steps in morning). Jones fracture (5th MT diaphysis, poor healing) vs avulsion of 5th MT tuberosity (fibularis brevis). March fracture (2nd MT). Lisfranc injury (tarsometatarsal).
- Sole: 4 layers; medial plantar nerve (≈ median) and lateral plantar nerve (≈ ulnar).

13. Lower limb nerves & spinal levels — MUST KNOW

Nerve	Injured by	Motor deficit	Sensory loss
Common fibular (L4–S2)	**Fibular neck fracture, tight cast, leg crossing, lateral knee trauma**	**Foot drop (↓ dorsiflexion), steppage gait, ↓ eversion; foot plantarflexed & inverted *("PED = Peroneal Everts and Dorsiflexes; if injured, foot dropPED")***	**Lateral leg, dorsum of foot**
Deep fibular	Anterior compartment syndrome	Foot drop (TA, EHL, EDL)	**First web space** only
Superficial fibular	Lateral leg trauma	↓ eversion	Dorsum of foot (spares 1st web space)

Tibial (L4–S3)	**Knee trauma, Baker cyst, tarsal tunnel**	**Can't plantarflex/invert, can't stand on toes; foot dorsiflexed & everted *("TIP = Tibial Inverts and Plantarflexes; if injured, can't stand on TIPtoes")***	**Sole of foot**
Femoral (L2–L4)	**Pelvic fracture, femoral catheterization, psoas hematoma**	**↓ knee extension, ↓ hip flexion; ↓ patellar reflex**	**Anterior thigh, medial leg (saphenous)**
Obturator (L2–L4)	**Pelvic surgery, labor, anterior hip dislocation**	**↓ thigh adduction**	**Medial thigh**
Superior gluteal (L4–S1)	IM injection, hip surgery	Trendelenburg sign (contralateral pelvic drop)	None
Inferior gluteal (L5–S2)	Posterior hip dislocation	Can't climb stairs / rise from seat	None
Sciatic (L4–S3)	**Posterior hip dislocation, misplaced IM injection**	**Hamstrings + all below-knee muscles**	**Most of leg and foot (not medial)**
Lateral femoral cutaneous (L2–3)	**Tight belts, obesity, pregnancy (at ASIS)**	**None**	**Meralgia paresthetica: anterolateral thigh**

- **Disc herniation: usually posterolateral at L4–L5 or L5–S1 → compresses the nerve root BELOW** (L4–5 disc → L5 root: weak dorsiflexion/big toe extension, dorsum of foot numb; L5–S1 → S1: ↓ Achilles reflex, lateral foot). Positive straight-leg raise.
- **Cauda equina syndrome**: saddle anesthesia, bowel/bladder dysfunction, bilateral sciatica → emergency.
- Lumbar puncture between L3–L4 or L4–L5 (iliac crest = L4 spinous process). Spinal cord ends L1–L2 in adults (conus medullaris).

14. Lower limb vessels & lymphatics

- External iliac → (under inguinal ligament) femoral → **profunda femoris** (medial & lateral circumflex femoral; perforators) → femoral passes adductor hiatus → **popliteal** → anterior tibial (→ dorsalis pedis) + tibioperoneal trunk (posterior tibial + fibular).
- Pulses: femoral (mid-inguinal point), popliteal (deep, knee flexed), posterior tibial (behind medial malleolus), **dorsalis pedis** (lateral to EHL tendon).
- **Great saphenous vein: anterior to medial malleolus → medial leg/thigh → joins femoral vein at the saphenous opening** (CABG graft; saphenous nerve runs with it below knee). Small saphenous: behind lateral malleolus → popliteal vein (with sural nerve).
- DVT: deep veins of calf/popliteal/femoral → PE. Varicose veins: incompetent valves of superficial/perforator veins.
- **Lymph: superficial inguinal nodes drain lower limb (superficial), skin below umbilicus, perineum, external genitalia (not testes), anal canal below pectinate line, lower vagina.** Popliteal nodes drain the lateral foot/posterolateral leg (small saphenous territory). Testes → para-aortic nodes.